App.js Component

In React, a **component** is a reusable, self-contained piece of the UI that can be rendered and managed independently. **App.js is a Component**

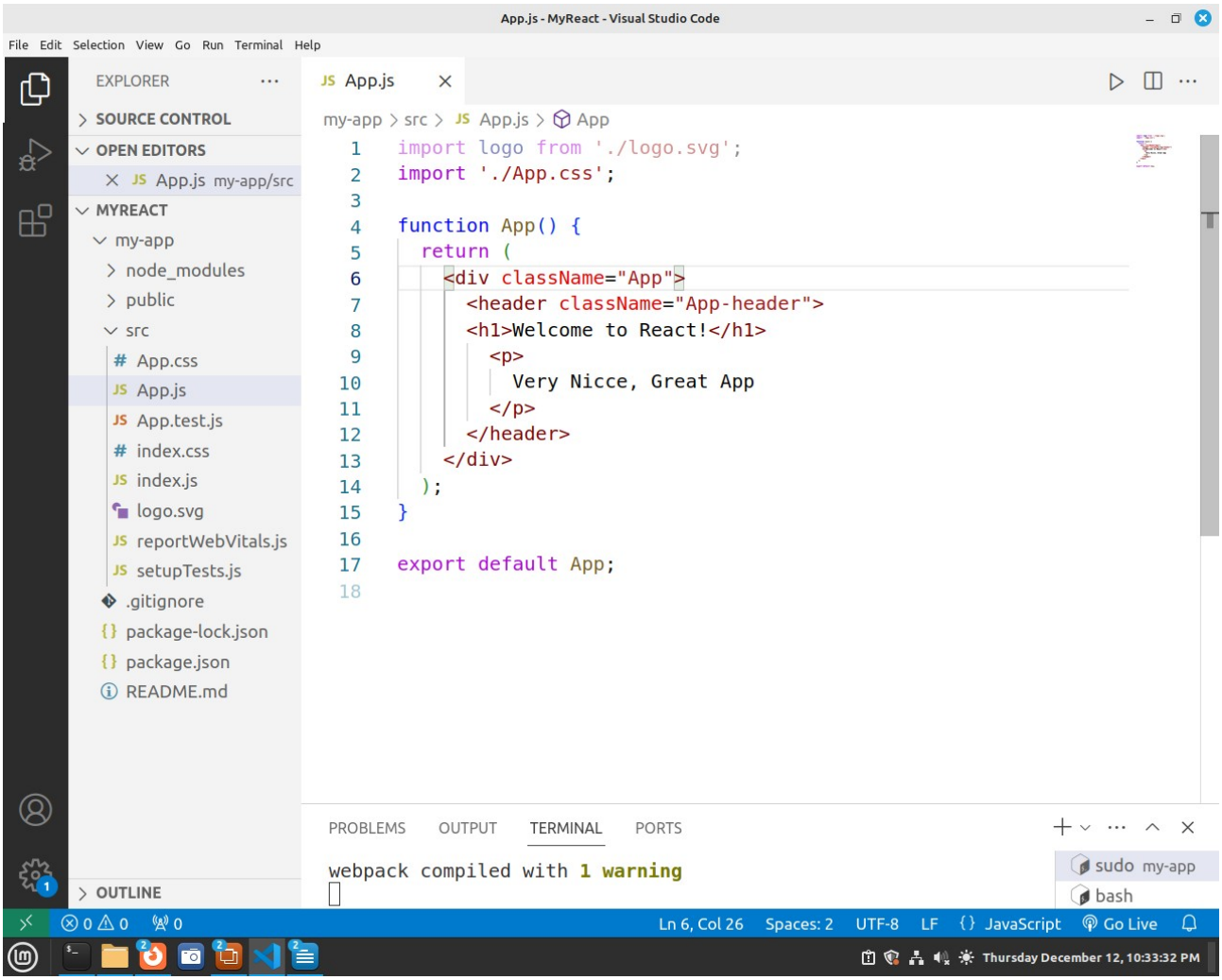
The App.js file is the central piece of a React application. It serves as the root component of your app, where the main structure, logic, and rendering occur. When you create a React app using tools like **Create React App (CRA)**, the App.js file is generated by default as the first component to start building upon.

Introduction to the App.js File

Here's a breakdown of what happens in a typical App.js file:

Default App.js Structure (from CRA)

When you create a new React app with create-react-app, the generated App.js file will look something like this:



1. **Importing React**

```
import React from 'react';
```

This line imports the React library, which is required for writing React components. Even though you may not always use React directly in the component code (because of JSX), it’s necessary for React's JSX syntax to work properly.

2. Importing Styles

```
import './App.css';
```

This line imports the App.css file, which contains the styles specific to the App.js component. By default, CRA sets up CSS files for styling

3. The App Component Function

Function Component: The App() function defines a functional component, which is one of the two primary ways to create components in React (the other being class components, though functional components are now preferred).

- **JSX (JavaScript XML)**: The return() block contains JSX, which allows you to write HTML-like code directly inside JavaScript. JSX will be converted into JavaScript calls by React behind the scenes.
- ** **: This div element wraps the entire content of the component. The **className="App"** is used instead of class, as class is a reserved keyword in JavaScript.
- **<header className="App-header">**: This defines the header section, with a class for styling.

4. Exporting the Component

```
export default App;
```

The App component is exported as the default export of the file. This allows other files (such as index.js) to import it and use it.

What Happens Next?

- **Render the Component**: The component gets rendered into the DOM through the index.js file. (Check index.js)

Rendering: The App component is inserted into the div with the id="root" in public/index.html. From there, React manages the DOM updates.

CSS - Cascading Style Sheet

We can include CSS styles in our JSX code. In **src**, check open a file named **App.css,**
This is the file we use to write our CSS, *This file is linked in App.js*
Lets change the Height of our App-header header class in App.js.

Open src/App.css and locate this CSS code.

```
.App-header {
  background-color: #282c34;
  min-height: 50vh;
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  font-size: calc(10px + 2vmin);
  color: white;
}
```

Change min-height : 50vh and background-color to blue

Run your App and see the style has changed.

Next, Give your heading a Class as shown below.

```
<h1 className='heading'>Welcome to React!</h1>
```

Then in App.css add below CSS code to style your heading class.

```
 /* Style heading class */
.heading {
    background-color: coral;
    font-family: 'Courier New', Courier, monospace;
}
```

Comments in react

In React, comments are similar to comments in regular JavaScript, but with some additional rules when working inside JSX (which is the syntax used to write UI components).

1. **JavaScript Comments (Outside JSX)**

When writing regular JavaScript in a React component (such as inside functions, variables, or logic), you can use standard JavaScript comments:

- **Single-line comment**:

  ```
  //This is a Comment
  ```

- **Multi-line comment:**

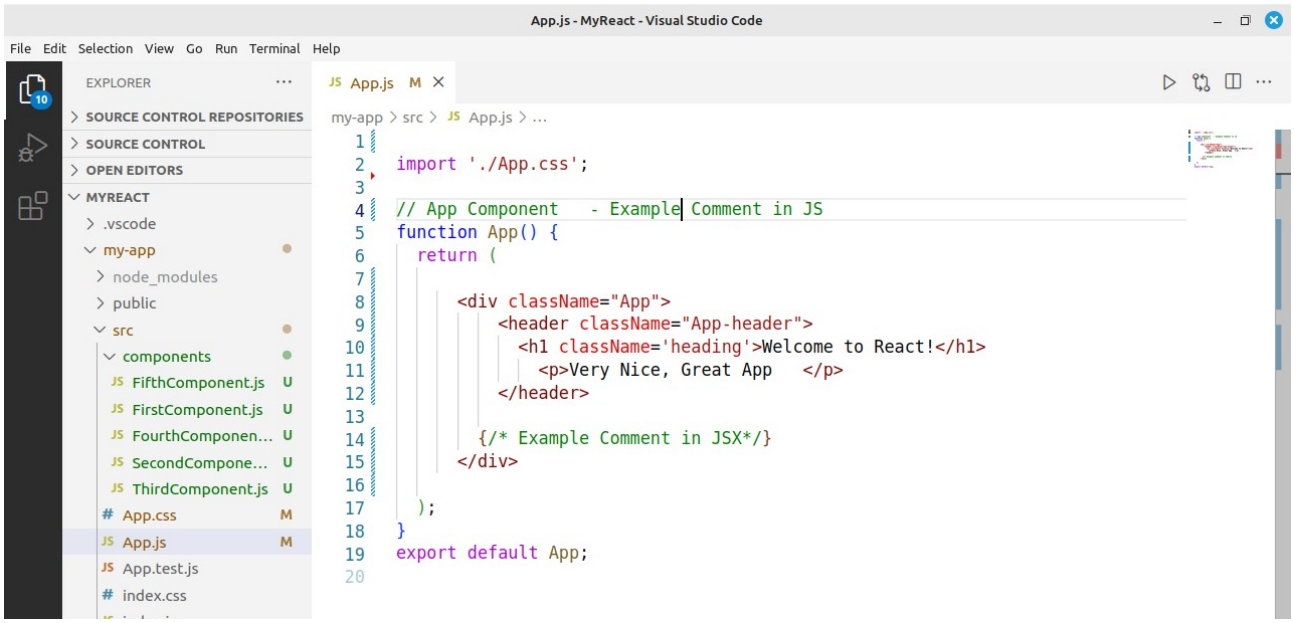
  ```
  /* This is a
  multi-line comment */
  ```

2. **Comments Inside JSX**

In JSX, you can't use regular JavaScript comments (// or /* */) directly inside the HTML-like syntax because it is parsed by JavaScript. Instead, you need to wrap the comment inside curly braces {} and use /* */ to comment inside JSX.

- **Single-line or multi-line comment inside JSX**:

```
<div>
  {/* This is a comment inside JSX */}
  <p>Some content</p>
</div>
```



Important Points:

- **Inside JSX**: Use {/* comment */} to wrap comments. JSX treats anything inside {} as JavaScript, so this is necessary to add comments.
- **Outside JSX**: You can use standard JavaScript comments (// or /* */).